



Graphs in Machine Learning

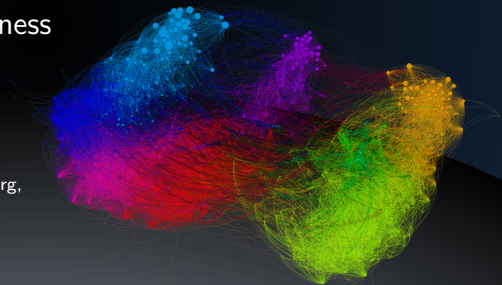
Graph Laplacian Basics

Definition, Properties, Smoothness

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Isara Labs

Partially based on material by: Ulrike von Luxburg,
Gary Miller, Doyle & Schnell, Daniel Spielman



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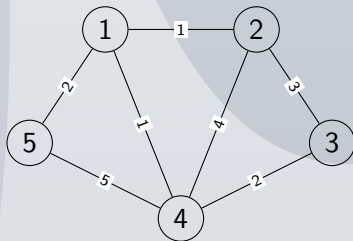
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L = D - W graph **Laplacian** matrix

$$\mathbf{L} = \begin{pmatrix} 4 & -1 & 0 & -1 & -2 \\ -1 & 8 & -3 & -4 & 0 \\ 0 & -3 & 5 & -2 & 0 \\ -1 & -4 & -2 & 12 & -5 \\ -2 & 0 & 0 & -5 & 7 \end{pmatrix}$$



L is SDD!

demo: <https://dominikschmidt.xyz/spectral-clustering-exp/>

Properties of Graph Laplacian

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Proof:

$$\begin{aligned} \mathbf{f}^\top \mathbf{L} \mathbf{f} &= \sum_{i,j \leq N} \mathbf{L}_{i,j} f_i f_j = \sum_{i,j \leq N} \mathbf{D}_{i,j} f_i f_j - \sum_{i,j \leq N} \mathbf{W}_{i,j} f_i f_j = \sum_{i=1}^N d_i f_i^2 - \sum_{i,j \leq N} w_{i,j} f_i f_j \\ &= \frac{1}{2} \left(\sum_{i=1}^N d_i f_i^2 - 2 \sum_{i,j \leq N} w_{i,j} f_i f_j + \sum_{j=1}^N d_j f_j^2 \right) = \frac{1}{2} \sum_{i,j \leq N} w_{i,j} (f_i - f_j)^2 \end{aligned}$$

Review: Eigenvalues and Eigenvectors

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For symmetric $\mathbf{M}$, the **multiplicity** of λ is the dimension of the space of eigenvectors corresponding to λ .

$N \times N$ symmetric matrix has N eigenvalues (w/ multiplicities).

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$\mathbf{Q}$ has eigenvectors in columns and $\mathbf{\Lambda}$ has eigenvalues on its diagonal.

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Right-multiplying $\mathbf{MQ} = \mathbf{Q}\mathbf{\Lambda}$ by $\mathbf{Q}^T$ we get the **eigendecomposition** of $\mathbf{M}$:

$$\mathbf{M} = \mathbf{MQQ}^T = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T \leftarrow \sum_i \lambda_i \mathbf{v}_i \mathbf{v}_i^T$$

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Self-edges do not change the value of $\mathbf{L}$.

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$$\mathbf{L} = \begin{bmatrix} \mathbf{L}_1 & & & \\ & \mathbf{L}_2 & & \\ & & \ddots & \\ & & & \mathbf{L}_k \end{bmatrix}$$

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For $\mathbf{L}_i$ $(0, \mathbf{1}_{|V_i|})$ is an eigenpair, hence the claim.

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<https://misovalko.github.io/mva-ml-graphs.html>